Machine Learning and its Applications

Today's goals

Module 1
Machine Learning Fundamentals I: Basic Concepts

Module 2
Machine Learning Fundamentals II: Model Training and Evaluation

Module 8 Introduction
- Introduction to Module 8's Study
 Video • 1 min
- What Does "Good" Mean in Machine Learning Models?
 Reading • 5 min

Module 8-1 ML Model Training Fundamentals
- ML Model Training Fundamentals: Parameters, Hyperparameters, and Loss Functions
 Video • 8 min
- ML Model Training Fundamentals: Loss Functions and Gradient Descent
 Video • 5 min
- ML Model Training Fundamentals: Gradient Descent
 Video • 10 min
- Materials for ML model training
 Reading • 10 min
- What Does It Actually Mean to "Train" a Machine Learning Model?
 Reading • 5 min

Module 8-2 ML Model Evaluation Fundamentals
- ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation
 Video • 13 min
- ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation
 Video • 5 min
- Materials on ML Model Evaluation Fundamentals
 Reading • 10 min
- When a Model Fails: Is It Bias, Variance, or Something Else?
 Reading • 5 min

Module 8-3 Summary of the ML Process
- Summary of the ML Process
 Video • 5 min
- Materials for ML Process
 Reading • 10 min
- From Data to Decisions: Reconstructing the Machine Learning Pipeline
 Reading • 5 min
- Module 8 Learning Coach
 Dialogue • 5 min

Module 8 Quiz
- Module 8 Quiz
 Practice Assignment • Grade: 100%

Module 8 Assignments
- Module 8 Assignment 1
 Graded App Item • 30 min
- Module 8 Assignment 2
 Graded App Item • 30 min
- CSV file for Module 8 Assignment
 Reading • 1 min
- **Module 8 Recap**
 Reading • 5 min

Module 3
Data Preparation

Module 4
Support Vector Machines

Module 5
Artificial Neural Networks

Module 8 Recap

0:00 / 1:30 2x

From Training to Trust: What Makes a Model Reliable?

Over this module, you moved beyond building machine learning models and focused on **how they are trained and evaluated**.

Reflect on the following:

1. Training is not learning

You saw that training a model is fundamentally an **optimization process** driven by a loss function.

- What exactly is being optimized during training, and what is *not* being optimized?
- Why does minimizing training loss alone fail to guarantee a reliable model?

2. Diagnosing model behavior

You learned to analyze models using **training error, test error, bias, and variance**.

Think of a model you trained (linear regression or kNN).

- What evidence would convince you that it is underfitting?
- What evidence would convince you that it is overfitting?

Why is generalization more important than performance on the training data?

3. Evaluation as a decision tool

Evaluation is not just reporting numbers—it guides decisions.

- How does cross-validation improve confidence in model comparison?
- Why can two models with similar average error behave very differently in practice?

4. Looking ahead to data preparation

In the next module, the focus shifts to **data preparation**.

- Based on what you learned about bias, variance, and generalization, why do you think data preparation is *not* a preprocessing detail but a core part of model quality?
- What data issues do you now expect could silently cause overfitting, underfitting, or misleading evaluation?

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